

ENTHUSIAST, LEADER & ACHIEVER COURSE

ALL PHASE

TARGET : PRE-MEDICAL 2020

Test Type : **MAJOR**

Test Pattern : **NEET(UG)**

TEST DATE : 05 - 04 - 2020

TEST SYLLABUS : FULL SYLLABUS

Q	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
A	2	3	2	2	1	4	1	2	2	4	1	3	3	3	1	2	1	4	2	3	2	3	3	2	1	1	3	3	3	2
Q	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
A	1	1	2	1	3	2	3	4	4	2	1	1	3	3	2	4	2	2	2	1	3	1	3	1	1	1	4	3	1	3
Q	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
A	1	1	1	4	1	4	3	3	4	4	4	3	2	3	4	1	1	3	2	3	2	2	4	4	1	4	1	4	3	1
Q	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120
A	4	3	4	1	1	1	3	2	4	3	4	4	4	4	3	2	4	4	3	3	2	3	4	2	3	4	1	1	3	2
Q	121	122	123	124	125	126	127	128	129	130	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A	3	3	4	4	2	1	4	3	2	2	4	1	4	3	3	4	2	1	3	2	1	2	2	1	3	4	2	3	2	2
Q	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170	171	172	173	174	175	176	177	178	179	180
A	4	3	2	3	4	3	3	3	1	2	4	4	1	3	1	3	2	4	2	3	1	1	2	3	1	4	2	2	4	2

HINT – SHEET

1. $\frac{1}{n}mg = \left(1 - \frac{1}{n}\right)\mu mg$

$\mu = \frac{1}{n-1}$

2. $a_{rel} = a_{real} - a_{frm} = g - a$

3. Avg. trans KE = $\frac{3}{2}nRT = \frac{3}{2} \times 5 \times \frac{25}{3} \times 300J$
= 18750 J

4. $\langle V \rangle = \frac{u \cos \theta t}{t} = u \cos \theta$

5. $|T_C - T_2| = 6 \text{ mg}$

6. $WD = \Delta K$

$= \frac{1}{2}m \left(\frac{v}{t} \right)^2$

7. $\frac{3}{4}mv_1^2 = mv_2^2$

$\frac{v_1}{v_2} = \sqrt{\frac{4}{3}}$

8. $\frac{t_1}{t_2} = \sqrt{\frac{a_2}{a_1}} = \sqrt{\frac{1}{1+1}}$

9. $g \propto R\rho$

$\frac{R_m}{R_e} = \frac{g_m \rho_e}{g_e \rho_m} = \frac{1}{6} \times \frac{5}{3}$

10. By energy conservation

$-\frac{GMm}{R} + \frac{1}{2}m \frac{1.5GM}{R} = -\frac{GMm}{R+h}$

$\frac{1}{4R} = \frac{1}{R+h} \Rightarrow h = 3R$

11. Force by spring $Kx = (m + M)a$

for $x = A = 4 \text{ cm}$, Maximum acceleration $a = \mu g$ to move together, $1000 \times 4/100 = (2+6) \mu \times 10$ on solving $\mu = 0.5$

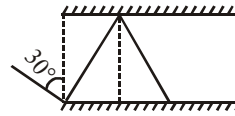
12. Sum of 2 periodic function is a periodic function with same period.

13. $\Delta \ell = \frac{FL}{AY} \propto \frac{L}{r^2}$

14. For maximum range hole must be in the middle of ground and liq. surface.

15. Energy liberated = $T (A_1 - A_2)$
 $= T(n4\pi r^2 - 4\pi R^2)$ and $(nr^3 = R^3)$
16. $5 = \frac{2\pi}{\lambda} \Rightarrow \lambda = \frac{2\pi}{5}$
17. $n\sqrt{T} = \text{constant}$
18. odd number of half loop $\left(\text{length } \frac{\lambda}{4}\right)$ are required for unison.
19. $\Delta n = n_i - n_f$
 $= \left(\frac{V}{V-v} - \frac{V}{V+v}\right)n = \frac{2nVv}{V^2 - v^2}$
20. The rate of coolings decreases as body cools.
21. $\therefore \frac{\Delta Q}{\Delta t} = \frac{\Delta \theta}{R} \Rightarrow \Delta t \propto R$
 As R becomes $\frac{1}{4}$ time becomes $\frac{1}{4}$
22. Let charge on smaller sphere be x
 $\frac{x}{4\pi r^2} = \frac{Q-x}{4\pi R^2} \Rightarrow x = \frac{Qr^2}{R^2 + r^2}$
 and $V = \frac{Kx}{r} + \frac{K(Q-x)}{R}$
23. $V_{A_1} = 0$
 Let $Q_{A_1} = x$
 $\frac{Kx}{r} + \frac{KQ}{R} = 0$
24. By conservation of charge
 $1500 + 100 = 2V_f$
 $V_f = 800 \text{ V}$
25. Accordingly $\frac{R_1 R_2}{R_1 + R_2} + R_3 = R_1$
26. Current in loop = 3 A
 $V_A - V_B = 3 + 2 - (1\Omega)(3 \text{ A}) = 2 \text{ V}$
27. $I = neAV_d$
 $V_d = \frac{I}{neA}$
28. $\frac{E_1}{E_2} = \frac{x_1}{x_2} \Rightarrow \frac{E_1}{3} = \frac{60}{45}$
 $E_1 = 4 \text{ V}$
30. $\sqrt{\frac{M^2}{4} + \frac{M^2}{4}} = \frac{M}{\sqrt{2}}$
31. $t_{1/2} = \frac{L}{R} \ln 2$

32. $Q = \frac{\phi_1 - \phi_2}{R} = \frac{Ba^2}{R}$
33. $I_{\max} = 1 \text{ A} \Rightarrow I_{\text{rms}} = \frac{1}{\sqrt{2}} \text{ A}$
 $I_{\text{WL}} = I_{\text{rms}} \sin \phi$
 $\Rightarrow \sin \phi = \frac{1}{\sqrt{2}} \Rightarrow \phi = 45^\circ$



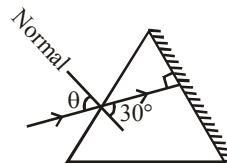
34. Base of triangle.
 $= 10 \times \frac{2}{\sqrt{3}} = \frac{20}{\sqrt{3}}$

Number of triangles = $\frac{2000\sqrt{3}}{20} = 100\sqrt{3}$

and 2 reflections every triangle.

35. Magnified real image is formed by concave mirror. Here $V = 3u$ (in same side)
 $|v - u| = |2u| = 80 \Rightarrow u = -40 \text{ cm}$
 $\frac{1}{f} = \frac{1}{v} + \frac{1}{u} = \frac{1}{3(-40)} + \frac{1}{-40} = \frac{1+3}{-120} = -\frac{1}{30}$
 $f = -30 \text{ cm}$

36. $\frac{h}{2\mu} + \frac{h/2}{1.5\mu} = \frac{h}{2} \Rightarrow \mu = \frac{5}{3}$



37. $\sin i = \sqrt{2} \frac{1}{2}$
 $i = 45^\circ$

38. $\frac{1}{10} + \frac{1}{-f} = \frac{1}{f} \Rightarrow \frac{1}{10} = \frac{2}{f}$

39. Independent light waves can never be coherent.

41. Remaining fraction = $(0.9)^4 = 0.6561$
 So decayed = $1 - 0.6561$

42. $\frac{\lambda_L}{\lambda_B} = \frac{\left(\left(\frac{1}{2}\right)^2 - \frac{1}{\infty}\right)}{\left(1^2 - \frac{1}{\infty}\right)} = \frac{1}{4}$

43. $\lambda \approx \frac{1240}{4} \approx 310 \text{ nm}$

44. $R = \frac{2-0.5}{5} \times 10^3 = 300 \Omega$

45. $\overline{A+B} + A.B = \overline{A+B.A.B} = (A+B).(\overline{A+B})$

$= A.\overline{A} + A.\overline{B} + B.\overline{A} + B.\overline{B} = A.\overline{B} + B.\overline{A}$

46. 2g of H_2 STP has volume 22.4 L
1g of H_2 at STP has volume 11.2 L
similarly

2g of He = 11.2 L

3g of NH_3 = 3.95 L

1g of CH_4 = 5.6 L

47. For 4f orbital $n = 4$ $l = 3$
nodes = $n - l$ i.e. 3 planer node = l
i.e. 3

Radial nodes = $n - l - 1 = 0$

49. 100 gm soln $\rightarrow$ 63 gm HNO_3

$\frac{100}{1.4} \text{ mL} \rightarrow 1 \text{ mole } HNO_3$

Molarity = $\frac{1}{\frac{100}{1.4} \times \frac{1}{1000}} = 14M$

50. $\alpha = \frac{D-d}{(n-1)d}$

$\alpha = \frac{D-d}{d}$

$d = \frac{D}{\alpha + 1}$

More α Less d

51. $NH_4^+ + H_2O \rightleftharpoons NH_4OH + H^+ : K_{eq} = K_h = K_{a(NH_4^+)}$

52. gram equivalent of $K_2Cr_2O_7$ = gram equivalent of X^{n+}

$(6 \times 10^{-3}) \times 6 = (5-n) \times 9 \times 10^{-3}$

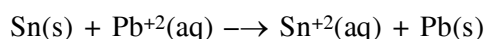
$5 - n = 4 \Rightarrow n = 1$

53. It shows octahedral voids (In fcc arrangement).

54. $P = 119X_A + 135$;

$\lim_{X_A \rightarrow 1} \frac{P}{X_A} = 119 + 135 = 254 \text{ torr}$

55. Cell reaction is :



Apply Nernst equation :

$E_{cell} = E_{cell}^0 - \frac{0.06}{2} \log \left[\frac{[Sn^{2+}]}{[Pb^{2+}]} \right] \dots (1)$

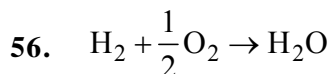
$E_{cell}^0 = -0.13 + 0.14 = 0.01 \text{ V}$

At equilibrium : $E_{cell} = 0$

Substituting in (1)

$0 = 0.01 - \frac{0.06}{2} \log \left[\frac{[Sn^{2+}]}{[Pb^{2+}]} \right]$

$\Rightarrow \frac{1}{3} = \log \left[\frac{[Sn^{2+}]}{[Pb^{2+}]} \right] \Rightarrow \left[\frac{[Sn^{2+}]}{[Pb^{2+}]} \right] = 2.15$



2g H_2 requires 16g O

So, 8g H_2 will require = 64 g O_2

But oxygen given is 40 g, so oxygen is a limiting reagent.

So, $\left(\frac{40}{32} \right) \times 4 \times 96500 = 5 \times t(\text{sec})$

$t = \frac{40 \times 4 \times 96500}{32 \times 5 \times 60} = 1608 \text{ min}$

57. $G = y\Omega^{-1}$; $k = x\Omega^{-1} \text{ cm}^{-1}$; $G = k \left(\frac{A}{\ell} \right)$

$\frac{\ell}{A} = \text{cell constant} = \frac{k}{G} \text{ cm}^{-1} = \frac{x}{y}$

Now, 0.1 M NaOH $\Rightarrow G = z = k \left(\frac{A}{\ell} \right)$;

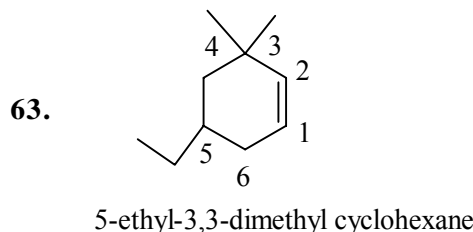
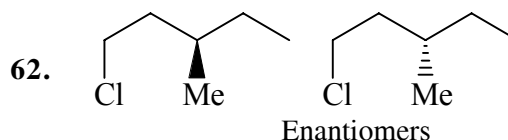
$k = z \times \left(\frac{\ell}{A} \right)$; $k = z \left(\frac{x}{y} \right)$

Molar conductivity = $\frac{k}{M} \times 1000 = \frac{\frac{zx}{y} \times 1000}{0.1}$

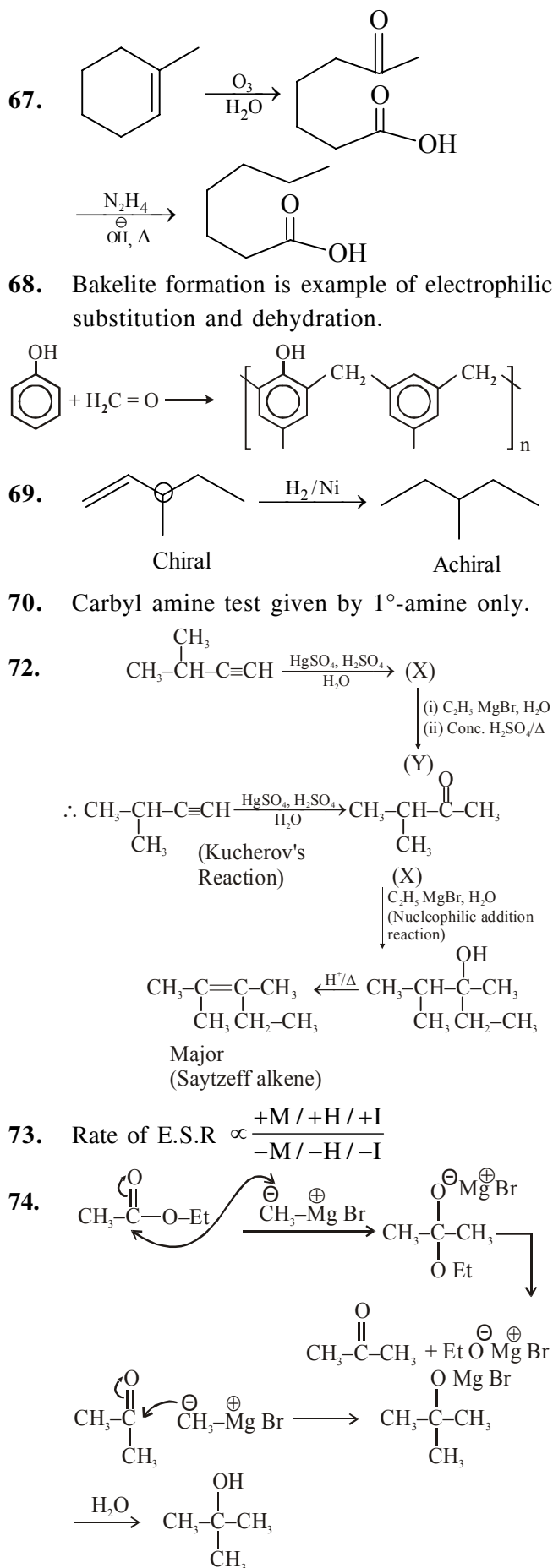
$= \frac{zx}{y} \times 10^4$

59. $\log k = \log A - \frac{E_a}{2.303R T}$
(y) (c) (m) (x)

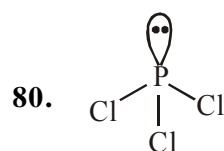
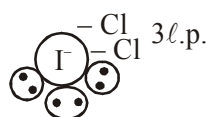
61. Order of acidity I > III > IV > II



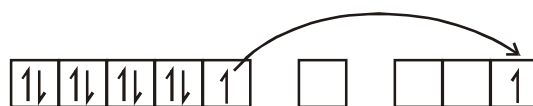
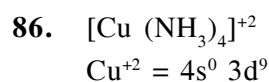
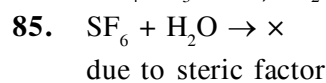
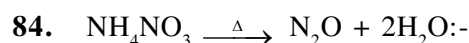
66. Stephen's reaction.



76. number of unpaired electron magnetic moment
- | | | |
|----------------------|---|----------|
| $\text{O}_2^\ominus$ | 1 | 1.73 B.M |
| $\text{O}_2^\oplus$ | 1 | 1.73 B.M |
| O_2 | 2 | 2.83 BM |
78. OCl_3
Side atoms different:-
79. ICl_2^-

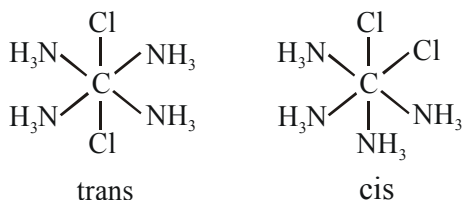


82. Based on property



dsp^2 , paramagnetic

83. $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]$ has 2 geometrical isomers



cis isomer has Cl-Co-Cl angle of 90°

88. $\text{M}(\text{AA})_2$ shows G.I.

